

Build night 3

Position Sizing and the Trade Ledger

Turn confidence into a capped size. Count overlapping bets once. Keep a ledger that adds up.

Travis Sim

NUS · Singapore

Thu 12 Nov 2026 · COM3/COM4

Tonight

18:00 to 20:00. Tonight's question: which sizing rule would you trust, and why?

18:00 — 10 min

Progress circle

last result and tonight's question

18:10 — 10 min

Mini-talk

overlapping bets, size from confidence

18:20 — 20 min

Demo tour

a ledger replayed, a refused order

18:40 — 20 min

Guided mission

size it, add one limit or cost, chart equity

19:00 — 50 min

Free build

your project; freeze main at 19:35

19:50 — 10 min

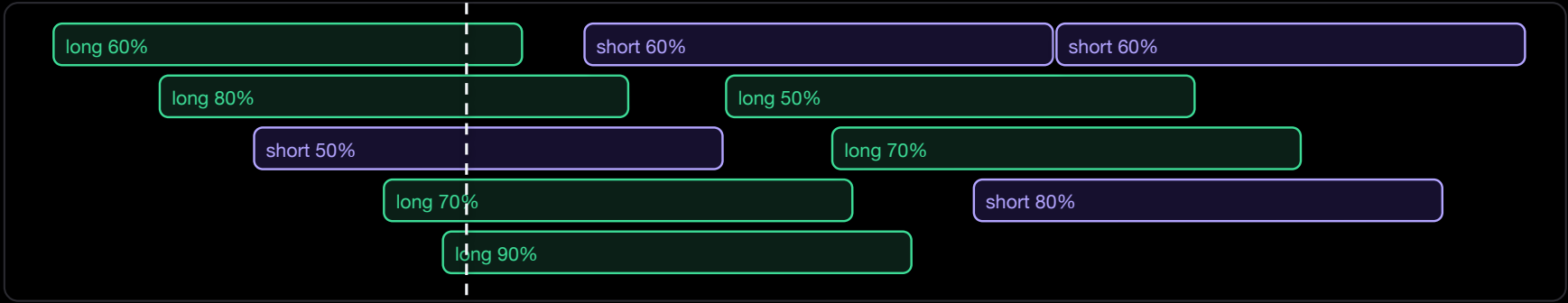
Show and tell

push, one line on next step

Open bets overlap

A bet starts at a signal and ends later. New signals keep arriving.

Each bar is one bet. Mint is long, lavender is short.



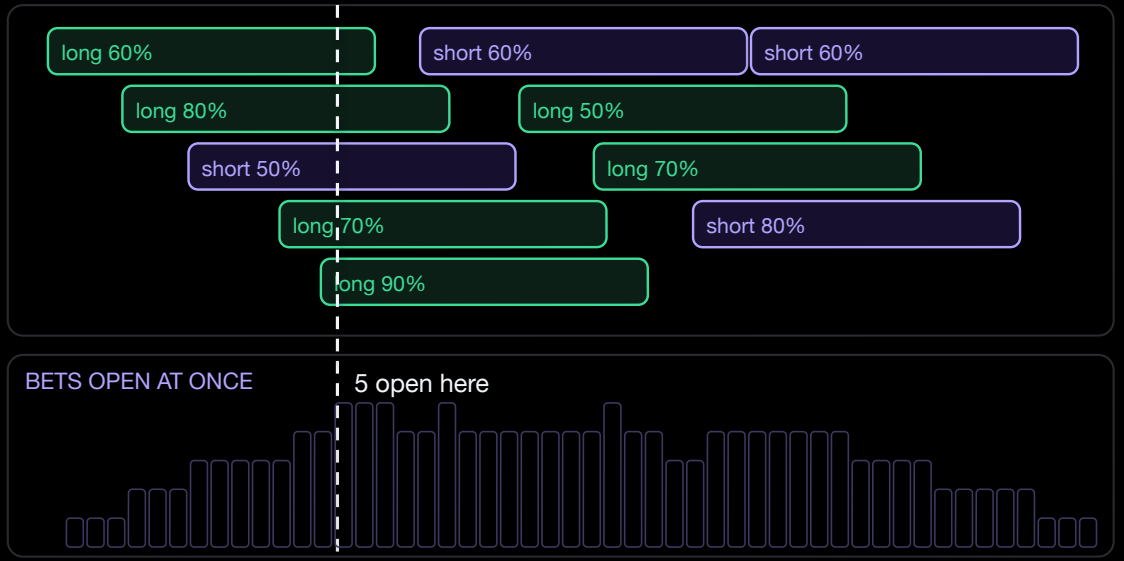
5 bets are open at the dashed line.

The same price move sits inside every bar that is open at that moment.

Illustrative bets, not real data

Count them together

Same bets, now with the number open at once.



ADD THEM UP

3 bets of 50% = 150%

Over the cap. One idea, counted three times.

AVERAGE THEM

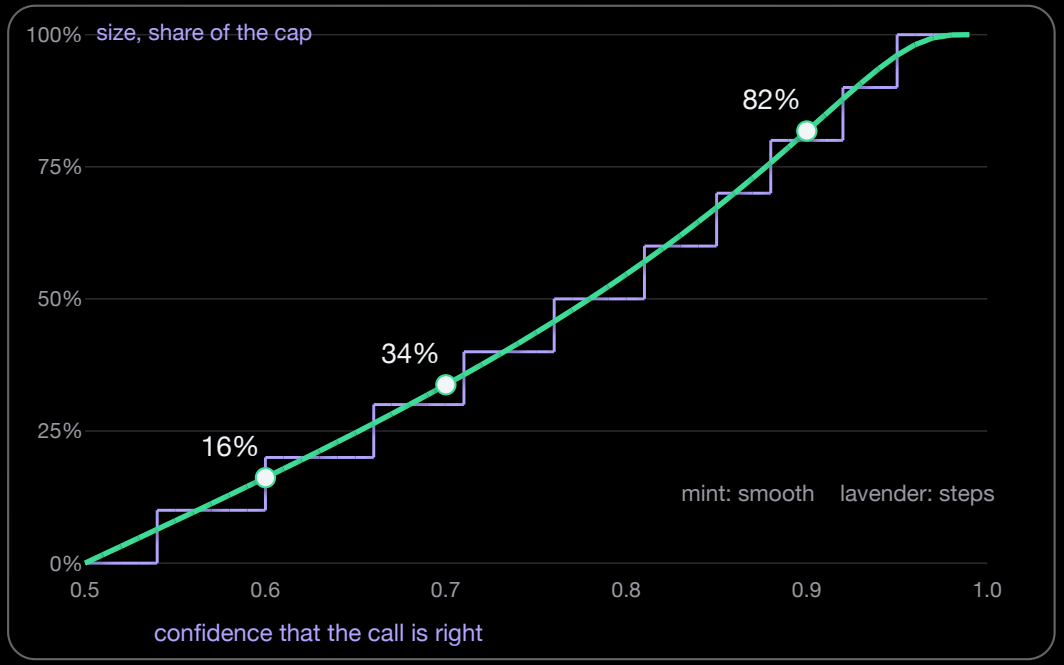
3 bets of 50% = 50%

One shared position. It moves when the bets change.

Illustrative bets, not real data

Size from confidence

A smooth curve. More confident, bigger. Never past 100%.



THE RULE

```
z = (p − 1/2) / sqrt(p(1−p))
size = 2 x Phi(z) − 1
```

Then round to steps of 0.1, so tiny changes do not trade.

The confidence in the sample code is a stand-in, not a calibrated probability.

Curve from code/out/sizing.json

Limits in order

The sizing proposes. Each check can shrink it or say no.

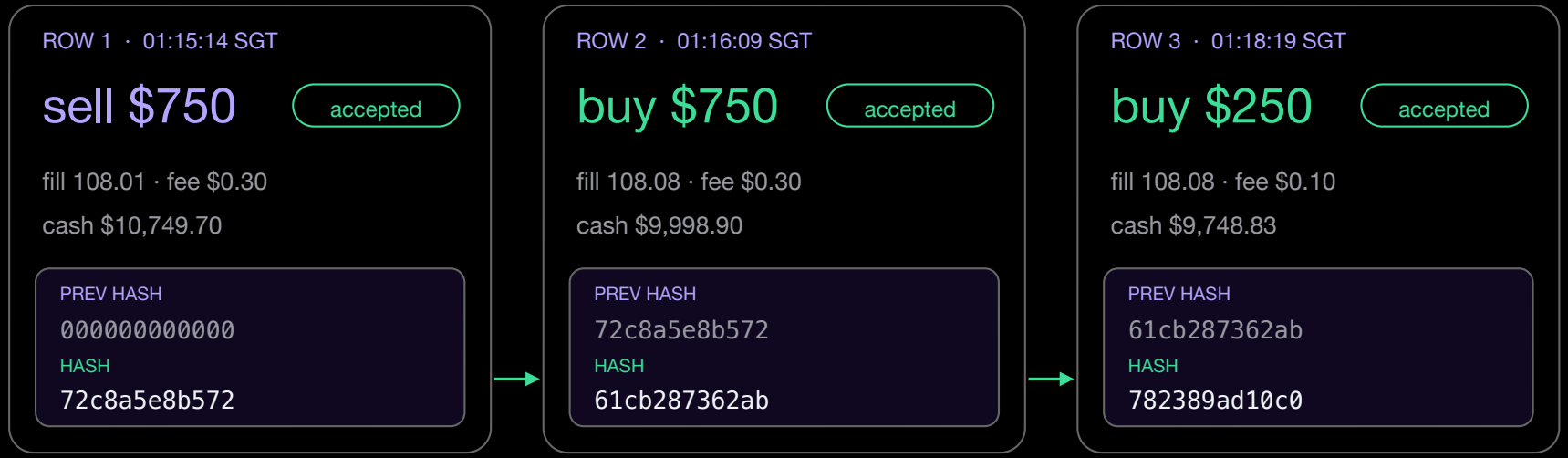


The hook is the last line. It says no, and it says why. (2000 USD exceeds the 1000 USD limit)

Numbers from code/out/sizing.json and code/src/risk.ts

One row per decision

Every order is a row, accepted or refused. Each row holds the hash of the row before it.



Append is the only way to change it. There is no edit and no delete.

The hash is a SHA-256 fingerprint of the row plus the previous hash. Shown: the first 12 of 64 characters.

Rows 1 to 3 of code/out/ledger.json

Edit a row and it shows

Replay the fills from the starting balance. If anything differs, the check names the row.

IT ADDS UP

Start	$10,000.00
Bought	-$5,498.62
Sold	+$5,000.75
Fees	-$4.20
Cash after replay	$9,497.93
Position	4.6058 SOL

Hash chain intact, balances match the fills (38 rows)

SOMEONE EDITS ROW 5

Row 5: the fill price changed by +$1.00

```
row 5: hash does not match the row's
contents (row was changed)
```

Hash chain BROKEN

Balances do NOT match the fills

The first problem is at row 5; later rows fail too.

Replay of code/out/ledger.json (recorded data); edit run with lab/shared/calc.js

Real numbers

Simulated, on recorded data; not advice.

BETS

37

momentum rule, 5 min each

MOST OPEN

7

4.3 on average

LEDGER ROWS

38

36 fills, 2 refused

FEES PAID

$4.20

4 bps per fill

EQUITY AT THE END, SIMULATED

no costs +0.024% → with costs -0.019%

average uniqueness 0.23; largest size averaged 38%, summed 100%

One 40-minute window, one night. A few dollars either way is noise, not a result.

Reads use public mainnet data; every write is devnet only; no real funds.

Source: code/out/ledger.json, concurrency.json, equity.json. Recorded Jupiter, Pyth and Orca prices, 2026-10-21 17:13 to 17:53 UTC (01:13 to 01:53 SGT). Simulated; not advice.

Real pool data

Inventory from real token accounts. Price impact from real Jupiter quotes.

ORCA SOL/USDC POOL HOLDS

178,736

SOL in its vault

10,692,375

USDC in its vault

A $1,000 order is 0.0094% of the USDC side.

PRICE MOVE FOR A BUY (JUPITER)

SOL/USDC

```
$100,000    →   0.005 bps
$1,000,000  →   0.066 bps
```

ORCA/USDC

```
$100,000    →   3.317 bps
$1,000,000  →   28.202 bps
```

LARGEST BUY UNDER 5 BPS

SOL/USDC

over $1,000,000

ORCA/USDC

about $159,000

SOL/USDC is deep. The impact cap only bites on thinner pools.

Source: code/recorded/ (Orca SOL/USDC pool vaults, public RPC; Jupiter quotes), captured 2026-10-21 19:36:08 UTC and 2026-10-21 19:36:02 UTC. Read with getAccountInfo; 1 bp = 0.01%.

The sizing in code

demo-repo/code/src/sizing.ts

```
export function probToSize(p: number): number {
  if (!Number.isFinite(p) || p <= 0.5) return 0;
  if (p >= 1) return 1;
  const z = (p - 0.5) / Math.sqrt(p * (1 - p));
  return 2 * normCdf(z) - 1;
}
export function discretize(m: number, step: number): number {
  if (!(step > 0)) throw new Error("step must be > 0");
  const r = Math.round(m / step) * step;
  return Math.max(-1, Math.min(1, Math.round(r * 1e9) / 1e9));
}
```

1 **Zero at a coin flip**

At or under 0.5 the size is 0. No edge, no bet.

2 **Smooth, then steps**

Phi is the normal curve. Steps stop tiny trades.

3 **No libraries**

The normal curve is a few lines in the same file.

Source: demo-repo/code/src/sizing.ts (excerpt: comments and blank lines removed)

Demo: the sizing curve

PREDICT FIRST

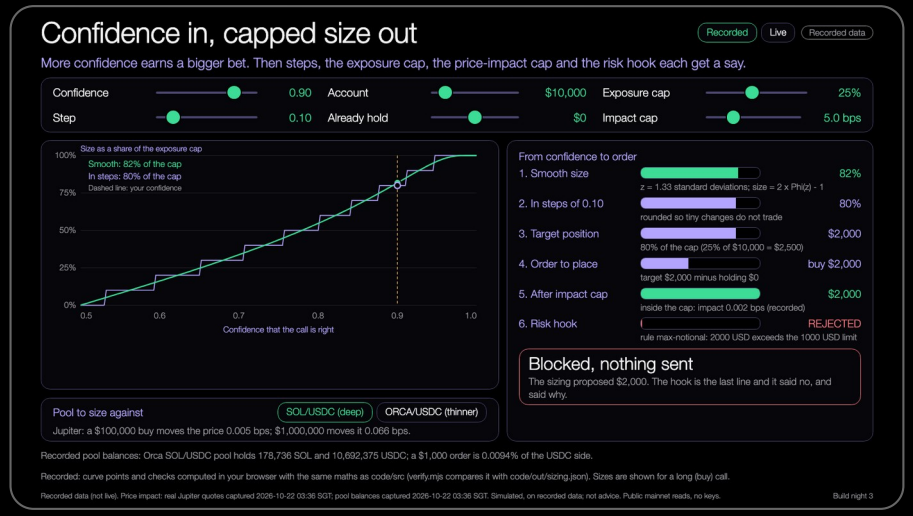
At what confidence does the risk hook start refusing the order?

```
$ npm run demo -- --recorded --max-fraction 0.25
```

YOU SHOULD SEE

- size rising with confidence, in steps
- the limit that bound, named

Fallback: Same recorded data. The curve is the same every run.



Learning Lab demo, runs from disk, recorded data.

Try it: nus-solana-demos.vercel.app/0005/01-sizing-curve.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo: overlapping bets

PREDICT FIRST

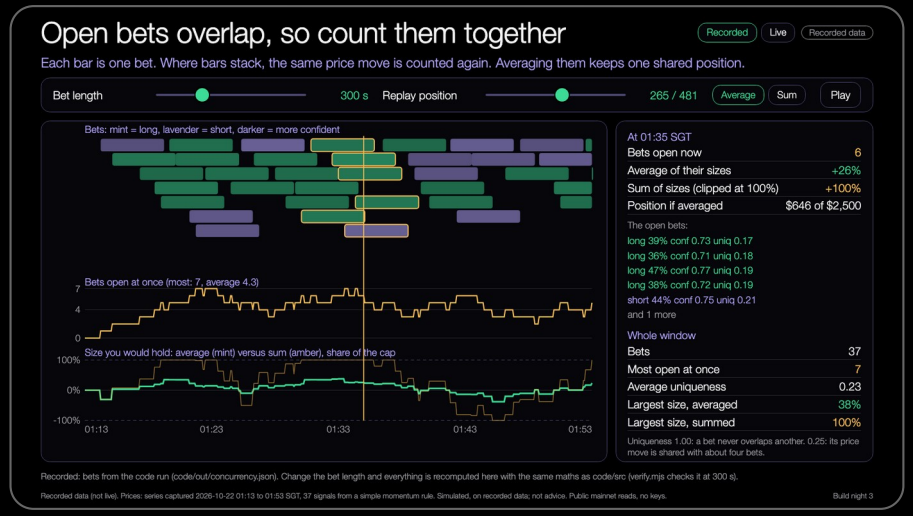
How many bets are open at the busiest moment?

```
$ npm run demo -- --recorded --horizon-sec 300
```

YOU SHOULD SEE

- bets stacked, with the count open at once
- average size versus summed size

Fallback: Same recorded data. The bets are the same every run.



Learning Lab demo, runs from disk, recorded data.

Try it: nus-solana-demos.vercel.app/0005/02-overlapping-bets.html

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo: a ledger replay

PREDICT FIRST

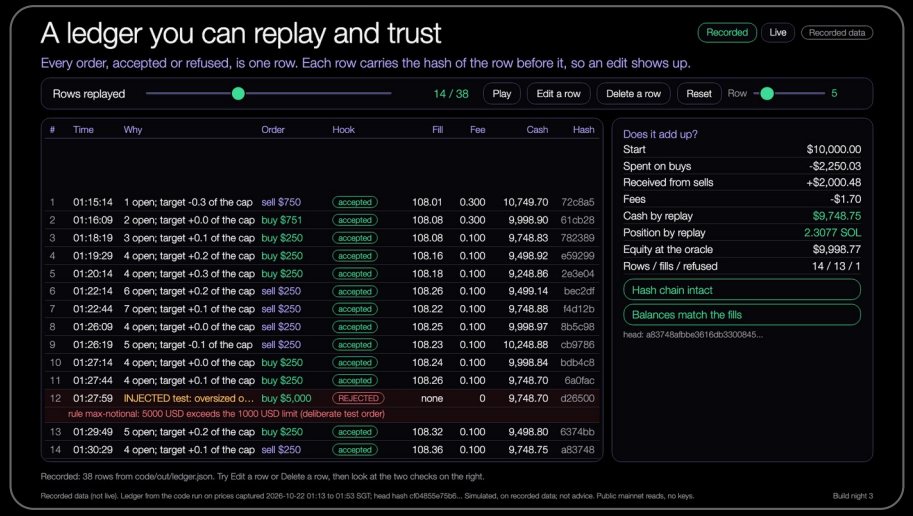
Edit row 5. Which check turns red, and which row is named?

```
$ npm run demo -- --recorded
```

YOU SHOULD SEE

- rows replayed, refusals with the rule and reason
- the hash chain and the balance check

Fallback: Same recorded ledger. The hashes are the same every run.



Learning Lab demo, runs from disk, recorded data.

`Try it: nus-solana-demos.vercel.app/0005/03-ledger-replay.html`

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Demo: equity before and after

PREDICT FIRST

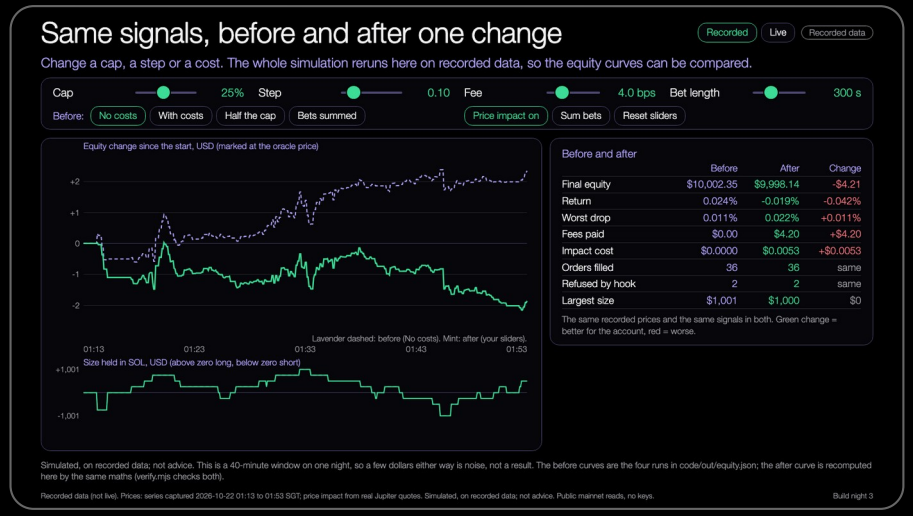
What does doubling the fee do to the final equity?

```
$ npm run demo -- --recorded --fee-bps 8
```

YOU SHOULD SEE

- before and after equity curves
- fees, refusals and worst drop side by side

Fallback: Same recorded data. Change one slider at a time.



Learning Lab demo, runs from disk, recorded data.

`Try it: nus-solana-demos.vercel.app/0005/04-equity-before-after.html`

Live mode: switch it on to watch real Solana data.

Source: demo-repo/lab/screenshots (captured from the Learning Lab demos on recorded data)

Try the demos yourself

DEMOS

`nus-solana-demos.vercel.app/0005/`

Open on your laptop. Move the sliders.

No install. The same files are in the repo and open from disk.



Scan to open the demos

Source: demo-repo/lab (the four Learning Lab demos, deployed as static pages, recorded data)

Your mission

18:40 to 19:00. Twenty minutes.

1

Run

The sample sizing on the sample ledger

2

Add

One limit or one cost

3

Chart

Equity, before and after

4

Check

The ledger still reconciles

Finish line: one chart with a before and an after.

Build menu

19:00 to 19:50. Pick one, or bring your own project.

BEGINNER	An append-only ledger for your project
BEGINNER	Add fees and impact to your ledger
STANDARD	Daily-loss and drawdown halts
STANDARD	Two sizing rules compared on one ledger
ADVANCED	Post your ledger's hash to devnet with a Memo

Before the break

19:35 to 19:50. Fifteen minutes.

1

Freeze

Commit a working main. Run it once from a clean clone.

2

Restart note

How to run it, what you should see, and your first task for January.

3

Devnet check

Leads confirm each team has devnet SOL before the break.

No work is expected over reading week, exams or vacation. Workshop 3 is Thu 21 Jan.

Three takeaways

1. Open bets overlap, so average them. Do not stack them.

2. Size grows with confidence, then caps and the risk hook have the last word.

3. A ledger is append-only and replays to the balance, refusals included.

Branch out

After tonight, in your own time.

BEGINNER

Run the sizing with a cap of 10% and then 40%. Compare the biggest size and the equity curve.

STANDARD

Add a daily-loss halt and write the halt into the ledger as a row.

ADVANCED

Compare two sizing rules on one ledger, or post the ledger's hash to devnet with a Memo.

Share back

Before you leave, and when we return.

TONIGHT

1

Freeze main and push your branch.

THU 21 JAN

2

Bring your restart note and ledger to Workshop 3.

ANY TIME

3

Send your repo link to the leads.

Get the code

REPO

`github.com/travisim/solana-agentic-trading-nus`

`folder: 0005-position-sizing-trade-ledger/demo`

```
1  $ npm install
```

```
2  $ npm run demo -- --recorded
```

Run both from the demo/code folder. Node 24. No keys, no AI credits. Recorded data is bundled.

Scan for this session's code

Let's connect



LinkedIn

linkedin.com/in/travisim



Telegram

t.me/travisim


SOLANA